

YONG FAH JIN

AI Engineer | Applied AI, Data Analytics & Automation | FinTech & IoT Monitoring

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PROFESSIONAL SUMMARY

AI Engineer-focused Software Engineer and Data Analyst with hands-on experience building full-stack AI products, integrating AI APIs, developing Python/JavaScript applications, and automating analytics workflows. Combines business operations, data analysis, IoT monitoring and software engineering experience to turn ambiguous workflows into practical AI-enabled tools. Recently completed Heicoders Academy GA100: Generative AI for Automation and Productivity and built Morning Kaki at the AI Engineer Hackathon. Experience includes ScribeRabbit, an AI video transcription and translation platform, a Toyota B2B recommendation engine, and scaling IoT monitoring from 40 to 500 devices.

CORE AI & TECHNICAL SKILLS

AI Engineering: LLM integration, generative AI APIs, prompt engineering, multimodal workflows, RAG/MCP prototyping, speech-to-text, text-to-speech, image generation

Programming & Web: Python, JavaScript, SQL, ReactJS, NextJS, VueJS, NodeJS/Express, FastAPI, Flask, REST APIs, microservices

Data, Cloud & Tools: pandas, numpy, scikit-learn, PostgreSQL, BigQuery, Grafana, GA/GTM, Hotjar, GCP, Azure, Kubernetes, Vercel, Git/GitHub, Linux

Delivery Strengths: AI product prototyping, observability, dashboard engineering, UAT, process mapping, stakeholder reporting, Agile delivery, technical documentation

SELECTED AI PROJECTS & HACKATHONS

Morning Kaki - *AI habit companion and PWA, AI Engineer Hackathon by 65labs, 2026* | morning-kaki.vercel.app/

- Built an AI-powered PWA that creates personalised morning nudges, using unique token-based setup and mobile push notifications after home-screen installation.
- Integrated LLM text generation, image generation, Whisper audio transcription and ElevenLabs text-to-speech to create a multimodal user flow during a 7-hour hackathon build.
- Designed for low-friction AI adoption by non-technical users, balancing practical habit tracking with a light-hearted user experience.

ScribeRabbit - *AI video transcription and translation platform, Sentient.io*

- Built a full-stack generative AI product using VueJS, Python FastAPI and GCP, integrating AI APIs to transcribe and translate uploaded videos end-to-end.

Toyota B2B Recommendation Engine - *collaborative filtering and API microservices, Sentient.io*

- Developed a recommendation engine that surfaced relevant parts based on similar customer behaviour, improving product discovery and cross-sell opportunities.

PROFESSIONAL EXPERIENCE

Remote Monitoring Analyst & Software Developer | **Terrenus Energy Pte Ltd** | Jan 2024 - Dec 2025

- Scaled IoT production monitoring from 40 to 500 solar devices, designing and maintaining 20+ Grafana dashboards for asset availability, alert volumes and incident resolution tracking.
- Reduced incident detection time from several days to under one hour by developing proactive offline-device alerting, reducing revenue-impacting downtime across the solar portfolio.
- Automated monthly stakeholder reporting with Python, pandas, numpy, PostgreSQL and SQL validation, removing manual data preparation and saving multiple hours per reporting cycle.
- Standardised alert definitions, KPI formulas, SOPs and monitoring workflows to reduce alert noise, improve onboarding and support finance and O&M reporting.
- Supported Azure and Kubernetes deployment activities and translated technical monitoring signals into weekly business summaries and actionable escalations.

Application Engineer (Contract) | **Nityo Infotech, Client: DBS Singapore** | Jan 2023 - Sep 2023

- Implemented production monitoring solutions using Python and observability tools to surface service health indicators and accelerate engineering incident response.
- Developed responsive internal web interfaces using ReactJS and Material-UI to support reporting workflows and day-to-day operational activities.
- Executed monitoring host upgrades and translated operational requirements into measurable metrics, alerts and dashboards for engineering stakeholders.

Application Engineer | Sentient.io Pte Ltd | Dec 2021 - Dec 2022

- Built and maintained ScribeRabbit, a full-stack generative AI transcription and translation product using VueJS, Python FastAPI, GCP and the company's AI APIs.
- Developed a Toyota B2B recommendation engine using collaborative filtering and REST API microservices to enhance product discovery and cross-sell opportunities.
- Conducted product analytics using BigQuery SQL and implemented GA/GTM/Hotjar tracking to surface funnel performance, user behaviour and conversion insights.
- Worked in sprint-based delivery cycles, translating direct client feedback into backlog items and iterative feature releases.

Business Support Analyst / Operation Analyst | Apple South Asia Pte Ltd | Aug 2015 - May 2021

- Led UAT cycles for internal tools including logistics stock allocation and sales CRM systems, mapping processes, documenting test cases and validating behaviour before rollout.
- Generated service level, staffing, ticket volume and KPI reports that supported leadership visibility, headcount discussions and resourcing decisions across JAPAC.
- Coordinated high-volume enterprise order fulfilment and NPI surge support across Retail, Logistics, Sales, Finance, Planning and external carrier partners.
- Maintained SOPs through multiple restructures and supported onboarding, off-boarding and operational issue resolution within the Regional Operations Center.

EDUCATION & CERTIFICATIONS

FlexiMasters in Financial Technology (FinTech) | Nanyang Technological University (NTU) | Expected Jul 2026

GA100: Generative AI for Automation and Productivity | Heicoders Academy | Completed 2026

Certificate in Software Engineering Immersive | General Assembly | Completed

Advanced Diploma in Python & ML; Certificate in Mobile App Development | Singapore Management University (SMU) | Completed

BSc Business, Second Upper Honours | SIM, University of London | Completed

ADDITIONAL INFORMATION

Citizenship: Singapore Citizen | **Availability:** Available immediately | **Languages:** English and Chinese